

ERIC SCHANET

SENIOR SOFTWARE ENGINEER

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PROFILE

Senior software engineer with 5+ years of experience building innovative, scalable and production-ready solutions for complex, quantitative problems. Significant expertise applying modern software engineering, DevOps and system design best-practices in a broad range of architectures and throughout all phases of the software development lifecycle. Core competencies include designing and building distributed systems, high-performance and cloud computing, predictive modelling using machine learning and stochastic methods, statistical data analysis, quantitative problem solving in cross-functional teams, and data visualization and interpretation. Experience working in large, international collaborations and leading projects from initiation to completion in an agile manner. Co-author of over 270 peer-reviewed scientific publications ([Google Scholar](#)).

EDUCATION

2021	Ph.D. Physics LMU Munich, CERN	Thesis: Searches for supersymmetry using data recorded at the LHC at CERN. Grade: <i>Summa cum laude</i> .
2018	M.Sc. Physics LMU Munich, CERN	Thesis: Multivariate statistical analysis of LHC data leveraging machine learning techniques. Grade: <i>Very good</i> .
2016	B.Sc. Physics LMU Munich, LFU Innsbruck	Thesis: Optimization studies on the search for gluinos and squarks. Grade: <i>Good</i> .

EXPERIENCE

Oct. 2021 – present	UBS Switzerland AG Senior Software Engineer Feb. 2024 – present Software Engineer Oct. 2021 – Jan. 2024	Technical lead for the migration of a business-owned risk and portfolio management platform into Azure Cloud. - Improved service scalability and uptime by migrating monolithic python application to microservice-based, API-first architecture using cloud-native resources like Kubernetes. - Designed, implemented and productionized first Kubernetes-based batch cluster within the organization. Development, operation and support of risk and portfolio management platform. - Significant business risk reduction by reducing the portfolio data pipeline lag from 3 to 1 business days through implementation of a data feed from the central mainframe. - Increased allocation and pledge structure transparency through implementation of a bespoke network-based asset allocation algorithm. - Improvements to the team’s efficiency, time to market and SLA through introduction of comprehensive CI/CD pipelines for all projects.
Nov. 2018 – Sep. 2021	LMU Munich, CERN Research Fellow	Novel method to approximate the statistical models of physics analyses. - Achieved 250x improvement in CPU performance for statistical inference. - Leveraged principal component analysis for optimizing efficiency and accuracy, utilized PyTorch tensors and implemented automatic differentiation of log-likelihood gradients. - Used in production for large-scale interpretations of 200+ million models.

Petabyte-scale statistical data analysis of CERN data using Monte Carlo models.

- Implemented containerized workflows for scalable *Data Analysis as a Service* using CERN Kubernetes cluster and GitLab CI/CD.
- Leveraged machine learning methods (RNNs, BDTs, clustering methods) for engineering new discriminative data features.
- Developed multivariate profile likelihood fit procedure, doubling the sensitive region of the analysis.

Low-latency grid computing monitoring and anomaly detection.

- Built ETL processes using raw data from CERN and WLCG databases, cleaning and consolidating into InfluxDB time-series and SQL databases.
- Implemented visualizations, dashboards and alerting using Grafana.
- Achieved 6x improvement in latency over alternative system, used in production by CERN grid computing operations.

S K I L L S

Programming Languages	Python, C/C++, Java, Rust, Go, JavaScript, Unix Shell Scripting, Objective-C, Swift, TeX, R
Libraries	NumPy, Pandas, Scikit-Learn, SciPy, Keras, PyTorch, Dask, FastAPI, pydantic, ReactJS
Databases	SQL (MySQL, SQLite), NoSQL (InfluxDB, MongoDB, ElasticSearch)
Data Visualization	Matplotlib, Seaborn, Flask, Grafana, Plotly, D3.js
Tools and Software	Docker, Kubernetes, CI/CD, Azure, AWS, Bicep, Terraform, OpenShift, ROOT
Operating Systems	Linux/Unix, MacOS, Windows
Version Control	Git, Subversion
Languages	English (fluent), German (fluent), French (fluent), Luxembourgish (native)

A C H I E V E M E N T S

March 2019

PhD Project Grant

Fellow of the Luxembourg National Research Fund

- Grant by the Luxembourg National Research Fund for research project searching for new physics using LHC data at CERN.

June 2019

Lindau Nobel Laureate Meetings

Fellow of the Wilhelm & Else Heraeus Foundation

- Participation in the 69th Lindau Nobel Laureate Meetings, where 580 selected young scientists met 39 Nobel Laureates in Physics.

E X T R A C U R R I C U L A R

Nov. 2018 – Sep. 2021

Particle Physics Masterclasses

- Speaker at regional and international masterclasses on particle physics, teaching students and high school teachers about the fundamental laws of nature.

Nov. 2011 – PRESENT

Astrophotography

- Photographs of planets, stars and galaxies in the night sky using telescope on custom-designed guiding mount. Published in journals and articles.

Oct. 2005 – Sep. 2021

Volleyball

- Active player and coach in Luxembourg and Germany. Team captain at CEV Small Countries Division Championship Finals with Luxembourgish national team.